Election Data Analysis Dashboard

End-to-End Data Analytics Project

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POWER BI | EXCEL | PYTHON | SQL

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Project Overview

This project analyzes election data to extract meaningful insights using a structured end-to-end pipeline. The data source is a **Kaggle dataset** (CSV/Excel format) containing constituency-level results, candidate information, party details, and voter turnout records across multiple election years.

Data Collection

Source Kaggle CSV/Excel election records

Data Cleaning

Remove nulls, duplicates, standardize names

Data Modeling

Build star schema for analysis

Dashboard Creation

Visualize insights in Power BI

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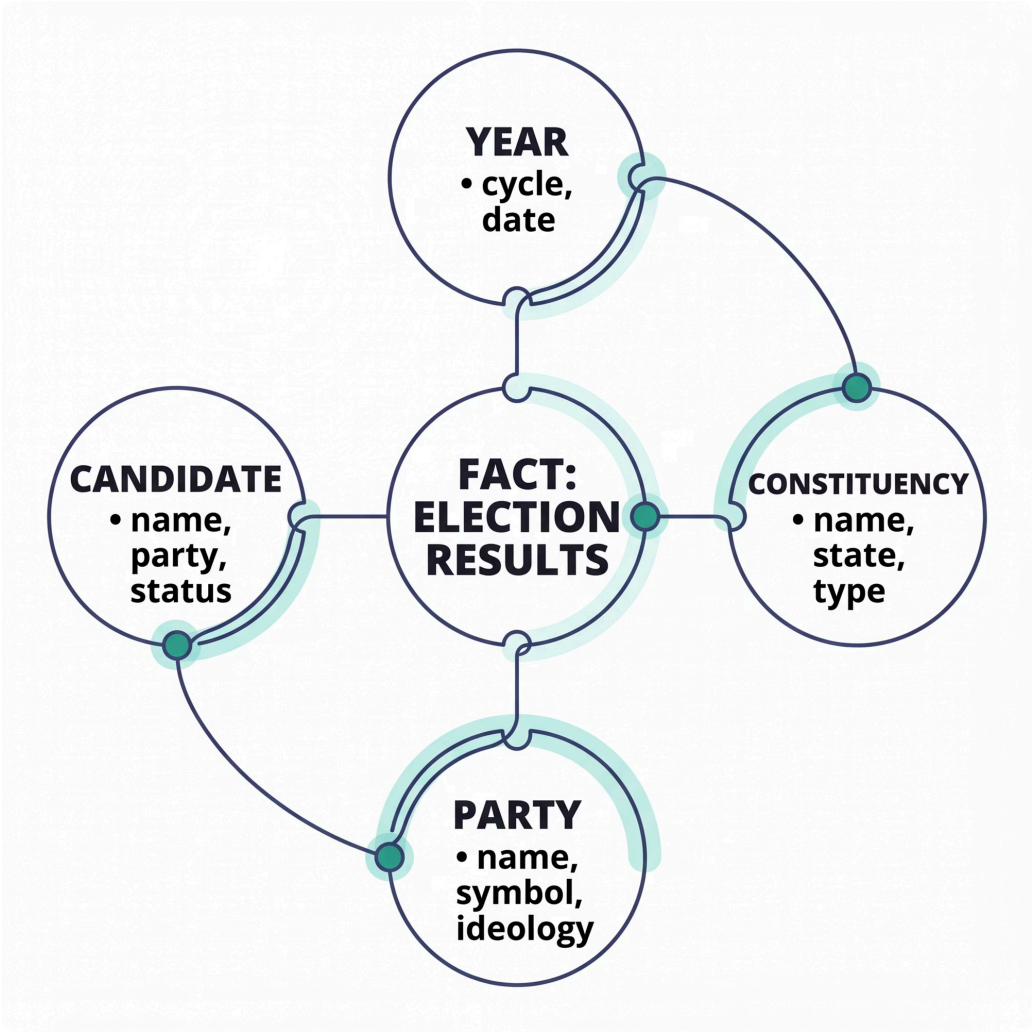
Data Integration & Modeling



Data Cleaning Pipeline

- Removed **null values** and duplicate records
- Standardized constituency and candidate names
- Normalized party names across election years

Star Schema Design

- **Fact Table:** Election Results (votes, margin, turnout)
- **Dimensions:** Candidate, Constituency, Party, Year

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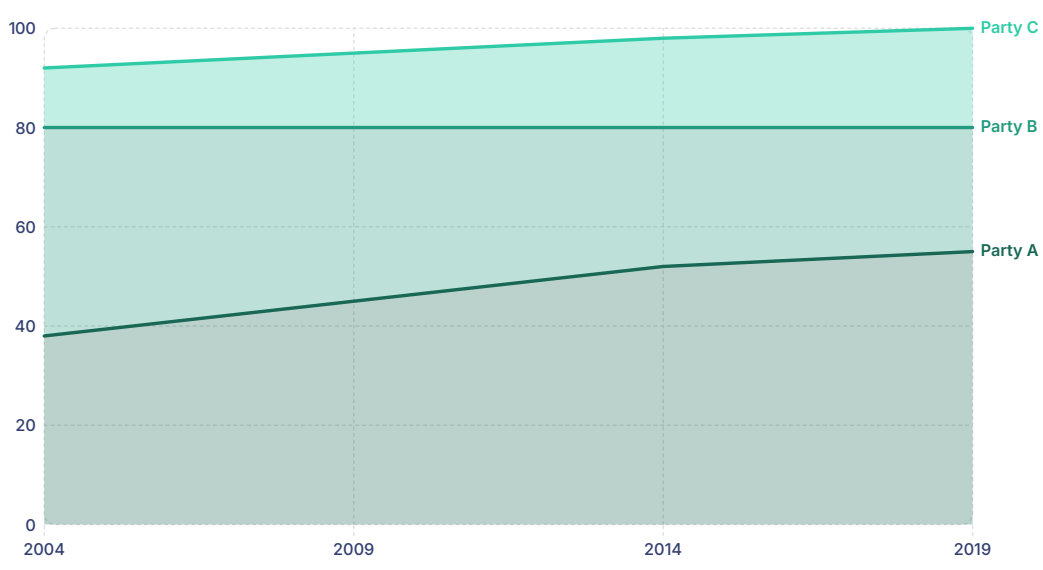
Historical Trends & Real-Time Dashboard



Vote Share Trends

Line and area charts reveal shifting party dominance and **swing constituencies** over time.

Dashboard Features

- Vote % bar charts with seat count indicators
- Filters: Year, Party, Region, Candidate
- Party strongholds and trend analysis

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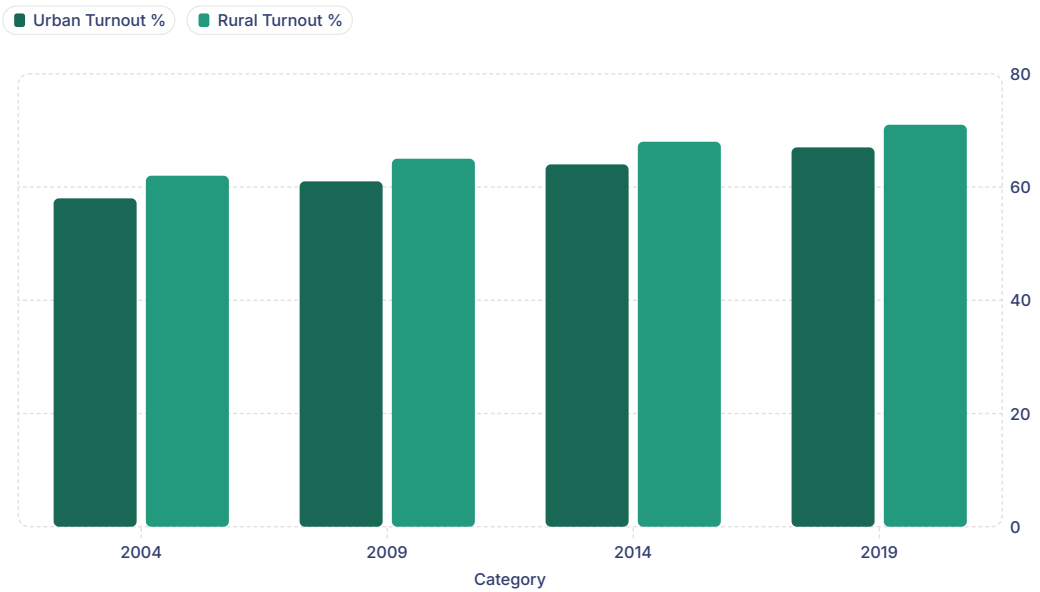
Turnout & Performance Analysis



Turnout Analysis

Breakdown by **gender, age group,** and geography reveals voting behavior patterns. Rural turnout consistently outpaces urban, with a narrowing gap over time.

Performance Tracking

- Winning margin analysis across constituencies
- Re-election and incumbent trends
- Party-wise seat share comparisons

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Conclusion & Insights

Key Findings

Identified party strongholds, turnout disparities by region and gender, and clear vote-share trends across election cycles.

Learning Outcomes

Mastered data cleaning, star schema modeling, and the critical role of visualization in communicating complex election data.

Future Scope

Integrate real-time APIs for live election tracking and apply predictive analytics to forecast outcomes and voter behavior.

Project by Rahul Gupta · B.Tech · Tools: Power BI, Excel, Python, SQL

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